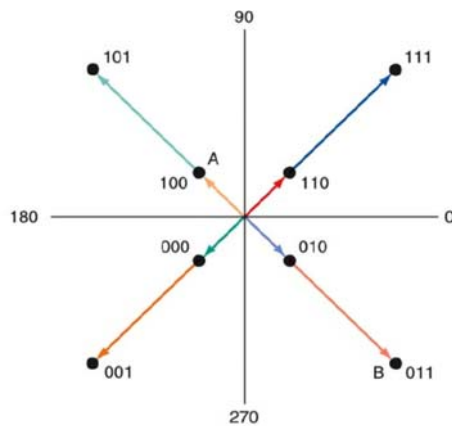


ENEL700 Tutorial 7

1. Explain the difference between FSK and FM (similarly for ASK vs AM, PSK vs. PM)

>> *FSK is a digital modulation technique while FM is an analogue modulation technique. In other words, only discrete frequency shifting is used in FSK. However, continuous frequency variation is allowed in FM.*

2. Draw the constellation diagram of 8-QAM.



>> (lecture notes, pg. 220)

3. Refer to slide 223 of lecture 7, explain the advantage and disadvantage of QPSK as compared to BPSK?

>> *We can increase the data rate (using one symbol to represent two bits) by using QPSK at the expense of higher E_b/N_0 if the same bit error rate is required.*

4. What is main difference between frequency hopping spread spectrum and direct sequence spread spectrum?

>> *Using pseudorandom sequence, FHSS scrambles the modulation carrier frequencies, while DSSS scrambles the data before modulation.*

5. Explain how orthogonal frequency division multiplexing (OFDM) works.

>> *Refer to the two figures in pp. 242-243 of lecture notes.*

ENEL700 Tutorial 8

1. Can we use a transmitter as a receiver? Justify your answers with simple sketches.

>> No, because the signal flows in a transmitter and in a receiver are in different directions. Refer to pg. 248 and pg. 255 of lecture notes.

2. An amplifier operating over the frequency range of 455 to 460kHz has a 200kΩ input resistor. What is the RMS noise voltage at the input to this amplifier if the ambient temperature is 17°C?

>> $v_N = \sqrt{4kTBR}$ $B=460-455=5\text{kHz}$, $R=200\text{k}\Omega$, $T=17+273.15=290.15\text{K}$, $k=1.38 \times 10^{-23}$, thus $v_N = \sqrt{4kTBR} = 4.0 \times 10^{-6}\text{V} = 4.0\mu\text{V}$

3. A receiver with a 75Ω input resistance operates at a temperature of 31°C. The received signal is at 89MHz with a bandwidth of 6MHz. The received signal voltage of 8.3μV is applied to an amplifier with a noise figure of 2.8dB. Find the output signal-to-noise ratio in dB.

>> input noise voltage, $v_N = \sqrt{4kTBR} = 2.749 \times 10^{-6}\text{V} = 2.749\mu\text{V}$

$$\text{Input signal to noise ratio, } SNR_{in} = \frac{v_{in}^2/R}{v_N^2/R} = \frac{(8.3 \times 10^{-6})^2}{(2.749 \times 10^{-6})^2} = 9.114$$

Noise factor of the amplifier, $F = \frac{SNR_{in}}{SNR_{out}}$, therefore

$$SNR_{out} = \frac{SNR_{in}}{F} = \frac{SNR_{in}}{10^{\frac{NF}{10}}} = \frac{9.114}{10^{\frac{2.8}{10}}} = 4.783 = 10 \log_{10} 4.783 \text{dB} = 6.797 \text{dB}$$

4. Calculate the noise figure (NF) of a cascade connections of amplifier and a mixer. The amplifier has a NF=5dB and a gain of 20dB, while the mixer stage has a NF=12dB and a gain of -2dB.

$$>> F = F_1 + \frac{F_2 - 1}{G_1} = 10^{5/10} + \frac{10^{12/10} - 1}{10^{20/10}} = 3.31 = 5.199 \text{dB}$$